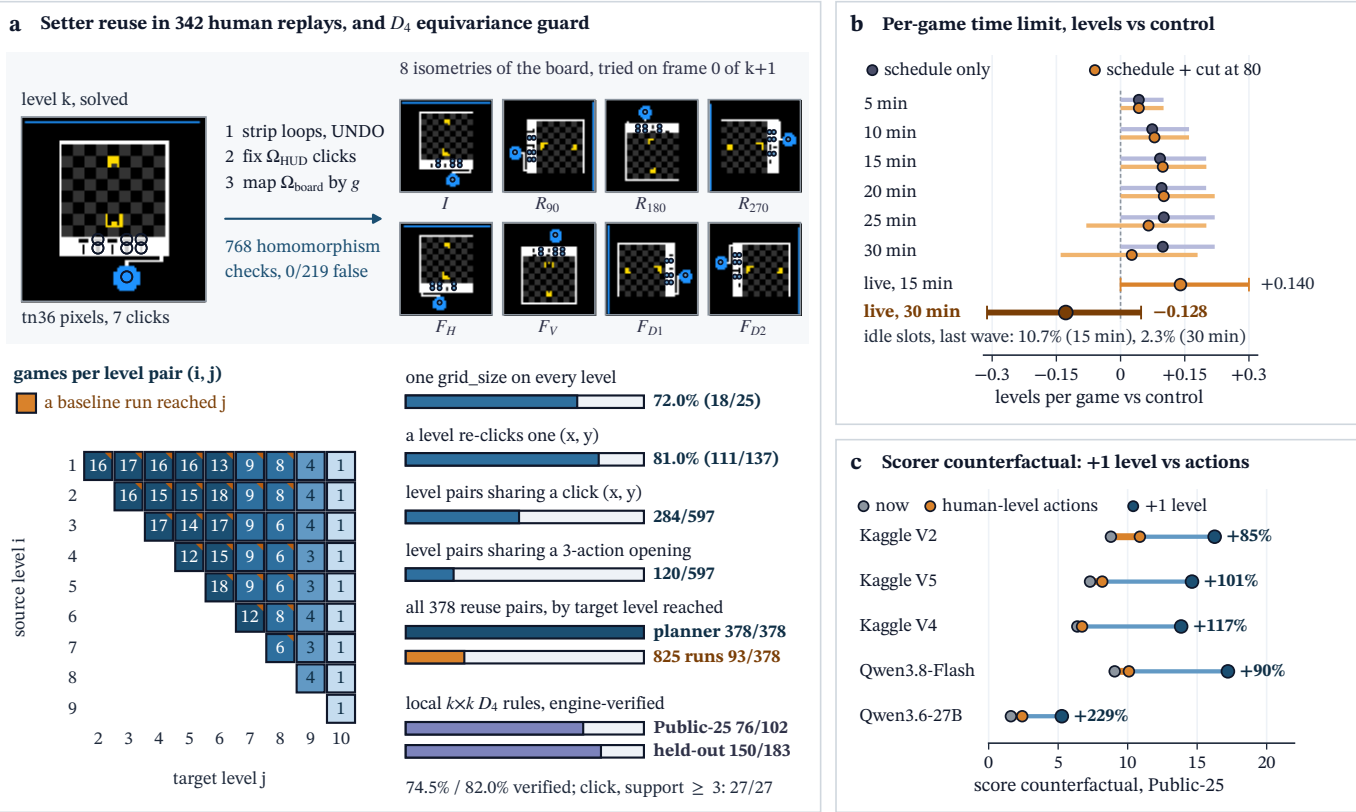

a Setter reuse in 342 human replays, and D_4 equivariance guard
8 isometries of the board, tried on frame 0 of k+1
level k, solved
tn36 pixels, 7 clicks
1 strip loops, UNDO
2 fix Ω_{HUD} clicks
3 map Ω_{board} by g
768 homomorphism checks, 0/219 false
I
R_{90}
R_{180}
R_{270}
F_H
F_V
F_{D1}
F_{D2}
games per level pair (i, j)
a baseline run reached j
source level i
target level j
one grid_size on every level
72.0% (18/25)
a level re-clicks one (x, y)
81.0% (111/137)
level pairs sharing a click (x, y)
284/597
level pairs sharing a 3-action opening
120/597
all 378 reuse pairs, by target level reached
planner 378/378
825 runs 93/378
local $k\times k$ D_4 rules, engine-verified
Public-25 76/102
held-out 150/183
74.5% / 82.0% verified; click, support ≥ 3: 27/25
b Per-game time limit, levels vs control
schedule only
schedule + cut at 80
5 min
10 min
15 min
20 min
25 min
30 min
live, 15 min
+0.140
live, 30 min
−0.128
idle slots, last wave: 10.7% (15 min), 2.3% (30 min)
levels per game vs control
c Scorer counterfactual: +1 level vs actions
now
human-level actions
+1 level
Kaggle V2
+85%
Kaggle V5
+101%
Kaggle V4
+117%
Qwen3.8-Flash
+90%
Qwen3.6-27B
+229%
score counterfactual, Public-25